

Santhosh Narayanan Baburaman

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EDUCATION

Master of Science in Analytics, University of Southern California (GPA: 3.92/4) Jan 2025 – Present
Relevant Coursework: Predictive Analytics, Optimization Methods for Analytics

Bachelor of Technology in Computer Science and Engineering, VIT (CGPA: 8.41/10) Sep 2020 – Apr 2024

EXPERIENCE

Machine Learning Intern – SmartInternz May 2023 – Jul 2023

- Built an end-to-end CNN-LSTM image captioning workflow on the Flickr8k dataset using structured preprocessing and hyperparameter tuning
- Developed a Flask-based prototype that generated captions for uploaded images, enabling interactive testing of model outputs
- Led a 3-member team coordinating model pipeline design, training iterations, and error analysis

Founder & President – ISE Analytics Student Association (IASA) Feb 2025 – Present

- Founded USC's ISE Analytics Student Association, hosting 12+ events for 500+ students and 20+ professionals

PROJECTS

eBay ML Challenge – Transformer-based NER (Python, PyTorch, NLP)

- Fine-tuned DistilBERT for Named Entity Recognition to extract product attributes (Brand, Model, Color) from eBay listings
- Trained and evaluated category-specific NER models with early stopping and macro-F1 tracking on 30K+ samples
- Automated text preprocessing, token alignment, and label mapping to support reproducible inference and submission

Reinforcement Learning Trading & Portfolio Optimization (Python)

- Formulated a finite-horizon Markov Decision Process with defined state, action, and reward spaces for trading decisions
- Implemented tabular Q-learning with Softmax exploration to learn execution and portfolio allocation policies
- Evaluated policy performance using cumulative reward, liquidation metrics, and policy behavior analysis

Automated Sales Analytics Copilot (Python, PostgreSQL, Claude API, Streamlit)

- Built an agentic LLM system (Claude API) with multi-step tool calling, text-to-SQL, and a 6-tool structured reasoning layer over a live PostgreSQL database
- Developed ARIMA + Prophet forecasting with 80% CI and a 4-rule statistical anomaly detection engine (z-score outliers, MoM trend reversals, margin compression)
- Deployed full-stack ML app on Streamlit with nightly GitHub Actions inference, automated Slack/email alerting, and a what-if scenario engine

NYC Taxi Demand & Fare Analysis (Python, Machine Learning, Time Series)

- Analyzed 1M+ NYC taxi trip records to model travel time, fare, and traffic speed using machine learning and time series approaches
- Engineered features and performed correlation analysis to identify key trip and temporal variables driving urban mobility patterns
- Co-authored an IEEE-published paper on predictive analytics for transportation and traffic forecasting

SKILLS

- **Programming & ML:** Python, PyTorch, Scikit-learn, Pandas, SQL
- **Machine Learning:** NLP, Transformer Models, Reinforcement Learning, ARIMA, Prophet, Predictive Modeling, Model Evaluation
- **Data & Systems:** PostgreSQL, Snowflake, Data Preprocessing, Feature Engineering
- **Cloud & Tools:** LLM APIs, GCP (Vertex AI, AutoML), Microsoft Azure, Git